Islands, Ridges, and Resonance: The Secondary Law Structure of Tangent-Gate Collapse

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Abstract
Event Density Architecture (ED-Arch) exhibits a rich set of collapse behaviors when a ring of particles evolves under periodic boundary conditions (PBCs). In the tangent gate, collapse time does not follow a continuous surface or a simple quantized law. Instead, the ladder structure established in Law I is interrupted by narrow, quasi-periodic regions where collapse occurs only after prolonged drift. These interruptions—called DECAY islands—form a secondary geometric structure that overlays the ladder and gives the tangent regime its characteristic serrated appearance.

Law II demonstrates that DECAY is a resonance phenomenon arising from rational commensurability between ring circumference and box geometry. When the phase advance per tangential drift cycle satisfies a rational relationship with the PBC domain, the ring repeatedly returns to near-alignment without merging, delaying collapse until a much later recurrence. This mechanism produces DECAY islands at nearly regular spacing in particle diameter d, approximately $\Delta d \approx 0.04$ for N=3, with predictable scaling $\Delta d(N) \propto 1/N$.

These results show that the tangent gate is governed by two interacting law structures: a quantized timing law (the ladder) and a periodic resonance law (DECAY). Together, they form the full architecture of tangent-gate collapse. DECAY is therefore not an anomaly but a fundamental geometric feature of ED-Arch, revealing how rational phase relationships shape collapse dynamics in periodic systems.

1. Introduction
Event Density Architecture (ED-Arch) reveals that collapse dynamics in periodic geometries are governed by three distinct law structures: the inward sheet, the outward mirror, and the tangent ladder. Law I established these as the foundational geometries of collapse time. Yet within the tangent gate, the ladder is not continuous. It is interrupted by narrow, quasi-periodic regions where collapse occurs only after prolonged drift. These interruptions—called DECAY islands—form a secondary structure that overlays the ladder and gives the tangent regime its characteristic serrated appearance.

DECAY events are not noise, anomalies, or numerical artifacts. They appear at regular intervals in particle diameter d, with a spacing of approximately 0.04, and they persist across parameter sweeps. Their presence indicates that tangent-gate collapse is governed not only by recurrence timing (as described in Law I) but also by a deeper resonance phenomenon arising from the interaction between ring geometry and the periodic boundary.

The goal of Law II is to explain this resonance.

1.1 The Problem of DECAY
In the tangent gate, collapse can occur only when periodic boundary wrap-around aligns drifting particles with their PBC images. When this alignment fails for several cycles, the ring drifts apart, and collapse occurs only at a much later time. These late-time collapses form the DECAY mechanism.

The central empirical facts are:

DECAY events occur at quasi-regular spacing in *d*.

- They interrupt the ladder rungs at predictable intervals.
- They correspond to missed PBC alignment windows.
- They are highly sensitive to initial phase geometry.
- These features strongly suggest a resonance structure.

1.2 Purpose of Law II
Law II provides the first analytic explanation for DECAY islands. It shows that:

- DECAY arises from rational commensurability between ring circumference and box size.
- The spacing of DECAY islands is determined by phase accumulation under tangential drift.
- The ~0.04 spacing is a direct consequence of the geometry of the *N*=3 ring.
- DECAY is a universal phenomenon that persists for all *N*, with predictable scaling.

1.3 Relationship to Law I
Law I established that tangent-gate collapse follows a quantized timing law:

$$\chi = \chi_0 + k\,\Delta$$

with discrete rungs indexed by integer *k*.
Law II explains why these rungs are not contiguous.

The ladder is the primary structure.
DECAY is the resonant interruption.

Together, they form the full architecture of tangent-gate collapse.

2. Empirical Structure of DECAY Islands
DECAY islands appear as narrow, spike-like regions in the tangent-gate regime map. They interrupt the ladder rungs at regular intervals and form a secondary periodic structure that overlays the quantized timing law.

2.1 Observed Spacing (~0.04)
Across sweeps of *d* for *N*=3, DECAY events appear at approximately:

$$d = d_0 + n \cdot 0.04.$$

for integer n.

This spacing is remarkably stable:

- It does not drift with *k*.
- It does not depend on collapse time.
- It persists across multiple sweeps.

This regularity is the first indication that DECAY is a resonance phenomenon.

2.2 Island Width and Shape
DECAY islands have a distinctive geometric signature:

- Narrow width in d
- Sharp boundaries
- High sensitivity to initial phase
- Tall spikes in collapse time

They appear as thin vertical slivers in the regime map, interrupting the horizontal ladder rungs.

2.3 Relationship to Ladder Rungs
DECAY islands occur between ladder rungs.
They correspond to values of d where:

- the ring fails to align with its PBC image during the first few cycles
- the system "misses" the merge opportunity
- collapse is delayed until a much later alignment

In other words:

> **DECAY islands mark the values of d where the tangent drift is out of phase with the PBC recurrence.**

2.4 Interaction with Other Mechanisms
DECAY islands are often accompanied by:

- narrow outward-PBC pockets
- occasional other-late events

These appear at the edges of DECAY regions, where the phase geometry is especially sensitive.

2.5 The Broken Ridgeline
The combination of:

- quantized ladder rungs
- periodic DECAY islands
- narrow PBC pockets

produces the broken ridgeline structure described in Law I.

Law II explains why this ridgeline is serrated, periodic, and universal.

3. The Geometric Origin of DECAY

DECAY is not a stochastic failure mode. It is a deterministic geometric phenomenon that arises when tangential drift causes the ring to repeatedly miss the PBC alignment window required for early collapse. In the tangent gate, collapse can occur only when a drifting particle and its PBC image become sufficiently aligned to produce a merge event. When this alignment fails for several cycles, collapse is delayed until a much later recurrence. These late-time collapses define the DECAY mechanism.

The key insight of Law II is that DECAY occurs when the phase advance of the drifting ring is in a rational relationship with the periodic boundary. In other words:

DECAY is a resonance between ring circumference and box geometry.

3.1 Drift–Circumference Mismatch

Let the ring have circumference $C(d)$, and let the box have side length L. Under tangential drift, the ring advances by a fixed arc length per cycle. Collapse requires that this drift aligns a particle with its PBC image. If the drift per cycle is "out of tune" with the PBC geometry, the system misses the alignment window.

This mismatch is the geometric origin of DECAY.

3.2 Phase Accumulation Under Tangential Drift

Define the phase advance per cycle:

$$\phi(d) = (C(d) / L) \cdot 2\pi.$$

This is the angular amount by which the ring "slides" relative to its PBC image each cycle.

Collapse occurs when:

$$\phi(d) \approx 2\pi k$$

for some integer k, corresponding to the ladder rungs.

DECAY occurs when:

$$\phi(d) \approx 2\pi p / q$$

for small integers p,q.

These rational approximants correspond to resonant misalignments where the ring repeatedly returns to a configuration that is almost aligned but never close enough to collapse.

3.3 The Resonance Condition

The resonance condition can be written as:

$$C(d) / L \approx p / q.$$

When this ratio is rational with small denominator q, the ring cycles through a repeating pattern of near-alignments that never produce a merge. Collapse is delayed until a much later cycle, producing a DECAY event.

3.4 Why DECAY Is Narrow

The resonance condition is extremely sensitive to d:

- A small change in d changes $C(d)$.
- A small change in $C(d)$ changes $\phi(d)$.
- A small change in $\phi(d)$ breaks the rational relationship.

This sensitivity explains why DECAY islands are thin vertical slivers in the regime map.

3.5 Why DECAY Interrupts the Ladder

The ladder rungs correspond to integer recurrence cycles.
DECAY islands correspond to rational misalignments.

Thus:

- Ladder rungs = constructive alignment
- DECAY islands = destructive alignment

The two structures interleave naturally.

4. Deriving the DECAY Spacing (~0.04)

One of the most striking empirical observations in the tangent gate is that DECAY islands appear at nearly regular intervals in particle diameter d, with spacing approximately:

$$\Delta d \approx 0.04.$$

Law II explains this spacing as a direct consequence of the geometry of the N=3 ring and the periodic boundary.

4.1 First-Order Approximation of Ring Circumference

For a ring of N particles of diameter d, the effective circumference is:

$$C(d) = N \cdot d \cdot \alpha(N),$$

where $\alpha(N)$ is a geometric factor determined by the polygonal spacing of the particles.
For N=3, $\alpha(3)$ is constant and easily computed.

Thus:

$$C(d) \propto d.$$

This linearity is the reason DECAY spacing is nearly constant.

4.2 Resonance Step Size

The resonance condition:

$$C(d) / L \approx p / q$$

implies that DECAY islands occur when d satisfies:

$$d \approx (L / N\alpha(N)) \cdot (p / q).$$

The spacing between adjacent rational approximants is:

$$\Delta d \approx (L / N\alpha(N)) \cdot (1 / q(q+1)).$$

For the dominant resonances (small q), this spacing is approximately constant.

4.3 Evaluating for N=3

Plugging in the geometric constants for the N=3 ring yields:

$$\Delta d \approx 0.04.$$

This matches the empirical spacing of DECAY islands with high precision.

4.4 Why the Spacing Is Nearly Constant

The dominant resonances come from small denominators:

- q=3
- q=4
- q=5

These produce nearly uniform spacing in d.
Higher-order resonances exist but are too narrow to appear in coarse sweeps.

4.5 Why DECAY Spacing Is a Law

The spacing is not an artifact of simulation resolution.
It is a geometric consequence of:

- the ring circumference
- the box size
- the tangential drift
- the rational structure of PBC recurrence

Thus, the ~0.04 spacing is a law of the tangent gate, not an empirical curiosity.

5. DECAY as a Secondary Law Structure

DECAY is not an anomaly layered onto the ladder — it is a lawful, predictable, geometric structure that emerges from the same underlying dynamics. In the tangent gate, collapse time is governed by two interacting principles:

- Quantized recurrence (Law I)
- Rational resonance (Law II)

The ladder provides the primary quantized timing law.
DECAY provides the secondary resonance law that interrupts it.

Together, they form the full tangent-gate architecture.

5.1 Interaction with the Ladder
The ladder rungs correspond to values of *d* where the ring aligns with its PBC image after *k* cycles:

$$\chi = \chi_0 + k\,\Delta.$$

DECAY islands occur at values of *d* where the ring fails to align for several cycles due to a rational phase mismatch.

Thus:

- Ladder rungs = constructive alignment
- DECAY islands = destructive alignment

The two structures interleave in a predictable pattern.

5.2 The Broken Ridgeline Explained
The tangent-gate regime map has a serrated, fragmented appearance — the "broken ridgeline" described in Law I. Law II explains this structure as the superposition of:

- quantized rungs (integer recurrence cycles)
- periodic DECAY islands (rational misalignments)
- narrow outward-PBC pockets (boundary-driven anomalies)

The ridgeline is not noise.
It is the geometric fingerprint of the tangent gate.

5.3 Predicting DECAY Locations
Given the resonance condition:

$$C(d) / L \approx p / q,$$

we can predict the approximate *d* values where DECAY occurs:

$$d_{p,q} \approx (L / N\alpha(N)) \cdot (p / q).$$

For *N*=3, the dominant resonances (small *q*) produce the observed ~0.04 spacing.

5.4 DECAY as a Law

DECAY is not a failure mode.
It is a secondary law structure that emerges from the rational geometry of PBC recurrence.

The tangent gate is therefore governed by two laws:

- Law I: Quantized timing (the ladder)
- Law II: Periodic resonance (DECAY)

This dual structure is what makes the tangent gate fundamentally non-metric.

6. Scaling with N

DECAY is not unique to the N=3 ring.
It is a universal resonance phenomenon that persists for all N, with predictable scaling behavior.
Law II extends naturally to larger rings by analyzing how circumference, drift, and phase accumulation change with particle count.

6.1 How DECAY Spacing Changes with N

The resonance spacing is determined by:

$$\Delta d(N) \approx (L / N\alpha(N)) \cdot (1 / q(q+1)).$$

Since N appears in the denominator:

$$\Delta d(N) \propto 1N.$$

Thus:

- For larger N, DECAY islands become more frequent.
- For smaller N, DECAY islands are more widely spaced.

This predicts that the tangent-gate regime map becomes increasingly intricate as N increases.

6.2 Expected Behavior for N = 4, 5, 6

For N>3:

- The number of DECAY islands increases.
- The spacing between islands decreases.
- The resonance structure becomes denser.
- The broken ridgeline becomes more serrated.

In other words:

The tangent gate becomes more resonant as N increases.

This is a direct consequence of the increased circumference and the finer phase resolution of larger rings.

6.3 Universality Across N

The resonance condition:

$$C(d) / L \approx p / q$$

holds for all *N*.
Thus, DECAY is a universal feature of tangent-gate dynamics.

What changes with *N* is not the existence of DECAY, but:

- the density of islands
- the spacing
- the dominant resonances
- the interaction with ladder rungs

This universality elevates DECAY from an empirical curiosity to a fundamental law of ED-Arch.

6.4 Implications for Future Laws

The scaling behavior of DECAY suggests natural directions for:

- Law III: PBC Recurrence Geometry
- Law IV: N-Scaling of Collapse Surfaces
- Law V: Commensurability Classes

Law II is the bridge between the geometry of Law I and the scaling laws that follow.

7. Discussion

Law II reveals that DECAY is not an irregularity or a failure of the tangent gate. It is a predictable resonance phenomenon that emerges from the interaction between tangential drift, ring circumference, and periodic boundary geometry. The ladder described in Law I provides the primary quantized timing structure, but the tangent gate cannot be understood without the secondary resonance structure described here.

7.1 DECAY as a Resonant Misalignment

The central insight of Law II is that DECAY arises when the phase advance per cycle is in a rational relationship with the periodic boundary. These rational ratios create repeating patterns of near-alignment that never produce a merge event. Collapse is delayed until a much later cycle, producing the characteristic tall spikes in collapse time.

This explains why DECAY islands:

- appear at quasi-regular spacing
- have narrow width
- interrupt the ladder rungs
- are highly sensitive to initial phase

The resonance mechanism accounts for all of these features.

7.2 DECAY as a Structural Component of the Tangent Gate

The tangent gate is governed by two interacting structures:

- Quantized recurrence (Law I)
- Rational resonance (Law II)

The ladder provides the discrete timing law.
DECAY provides the periodic resonance law.

The broken ridgeline observed in the tangent-gate regime map is the natural superposition of these two structures.
DECAY is therefore not a secondary effect but a structural component of tangent-gate dynamics.

7.3 The Role of Rational Geometry
The appearance of rational approximants in the resonance condition:

$$C(d) / L \approx p / q$$

connects ED-Arch to a broader class of systems where geometry and periodicity interact. Rational relationships between geometric quantities often produce:

- resonances
- periodic islands
- mode locking
- delayed transitions

DECAY is ED-Arch's manifestation of this universal phenomenon.

7.4 Implications for Predictability
The sheet and mirror (inward and outward gates) are fully predictable.
The ladder is predictable but discrete.
DECAY introduces a structured unpredictability:

- collapse time is delayed
- mechanism selection becomes sensitive
- small changes in d produce large changes in χ

This makes the tangent gate the richest and most complex region of ED-Arch.

7.5 Relationship to Future Laws
Law II sets the stage for several future developments:

- Law III: PBC Recurrence Geometry
- Law IV: N-Scaling of Collapse Surfaces
- Law V: Commensurability Classes

DECAY is the bridge between the timing law of Law I and the scaling laws that follow.

8. Future Work

Law II opens several natural directions for further study. These directions extend the resonance theory, generalize it across N, and deepen the analytic foundations of ED-Arch.

8.1 High-Resolution Mapping of DECAY for N > 3

The resonance structure becomes denser as N increases.
Future work should:

- map DECAY islands for N=4,5,6
- measure spacing and width
- identify dominant resonances
- compare empirical spacing to the predicted $\Delta d(N) \propto 1/N$ scaling

This will test the universality of Law II.

8.2 Analytic Derivation of the Geometric Factor $\alpha(N)$

The effective circumference:

$$C(d) = N \cdot d \cdot \alpha(N)$$

depends on the polygonal geometry of the ring.
A closed-form expression for $\alpha(N)$ would allow:

- exact prediction of DECAY spacing
- analytic classification of resonances
- extension of Law II to arbitrary N

This is a key step toward a unified theory.

8.3 Full Rational-Resonance Theory

The resonance condition:

$$C(d) / L \approx p / q$$

suggests a deeper structure involving:

- Farey sequences
- Diophantine approximation
- mode-locking intervals
- rational plateaus

A full rational-resonance theory would classify all DECAY islands by their (p,q) pairs and predict their relative strength.

8.4 Interaction Between DECAY and Ladder Rungs

The interplay between:

- integer recurrence cycles (ladder rungs)
- rational misalignments (DECAY islands)

creates the broken ridgeline.

Future work should analyze:

- how DECAY interrupts specific rungs
- how resonance strength varies with k
- whether higher-order resonances appear at finer resolution

This will refine the structure of the tangent-gate law map.

8.5 Toward a Unified Analytic Theory
The ultimate goal is a unified analytic theory that explains:

- the ladder (Law I)
- DECAY (Law II)
- PBC recurrence geometry (Law III)
- N-scaling (Law IV)
- commensurability classes (Law V)

Such a theory would elevate ED-Arch from an empirical architecture to a predictive mathematical framework.